Status	Finished
Started	Friday, 11 September 2026, 10:35 AM
Completed	Friday, 11 September 2026, 11:20 AM
Duration	44 mins 18 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Assume that the given string has enough memory.

Don't use any extra space(IN-PLACE)

Sample Input 1

a2b4c6

Sample Output 1

aabbbbccccc

Answer: (penalty regime: 0 %)

```
import sys


def solve():
  input_data = sys.stdin.read().split()
  if not input_data:
    return

  for s_str in input_data:
    # Convert string to a mutable list of characters, assuming enough extra memory (pad
    s_list = list(s_str)
    s_list.extend([''] * 200)

    orig_len = len(s_str)

    # Step 1: Calculate the final expanded length (handling multi-digit numbers)
    final_len = 0
    i = 0
    while i < orig_len:
      i += 1  # Skip the character
      count = 0
      while i < orig_len and s_list[i].isdigit():
        count = count * 10 + int(s_list[i])
        i += 1
      final_len += count

    # Step 2: Expand from right to left to prevent data overwriting
    read_idx = orig_len - 1
    write_idx = final_len - 1

    while read_idx >= 0:
      # Extract the numeric count from right to left
      count = 0
      base = 1
      while read_idx >= 0 and s_list[read_idx].isdigit():
        count += int(s_list[read_idx]) * base
        base *= 10
        read_idx -= 1

      # Extract the character
      if read_idx >= 0:
        ch = s_list[read_idx]
        read_idx -= 1

        # Write the character 'count' times backwards
        for _ in range(count):
          s_list[write_idx] = ch
          write_idx -= 1

    # Print the resulting expanded string
    print(''.join(s_list[:final_len]))
```

```
51 |    print(''.join(s_list[:final_len]))
52
```

	Input	Expected	Got	
✓	a2b4c6	aabbbbccccccc	aabbbbccccccc	✓
✓	a12b3d4	aaaaaaaaaaaabbbddddd	aaaaaaaaaaaabbbddddd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Write a Python program to get one string and reverses a string. The input string is given as an array of characters `char[]`.

You may assume all the characters consist of printable ascii characters.

Example 1:

```
Input:
hello
Output:
olleh
```

Example 2:

```
Input:
Hannah
Output:
hannaH
```

Answer: (penalty regime: 0 %)

```
import sys


def reverse_string():
  # Read the input string from standard input
  input_data = sys.stdin.read().split()
  if not input_data:
    return

  for s_str in input_data:
    # Convert the string into an array of characters (char[])
    s = list(s_str)

    # Reverse the array in-place using two pointers
    left, right = 0, len(s) - 1
    while left < right:
      s[left], s[right] = s[right], s[left]
      left += 1
      right -= 1

    # Print the reversed character array as a string
    print("".join(s))


if __name__ == "__main__":
  reverse_string()

```

	Input	Expected	Got	
✓	hello	olleh	olleh	✓
✓	Hannah	hannaH	hannaH	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Given a **non-empty** string s and an abbreviation abbr, return whether the string matches with the given abbreviation.

A string such as "word" contains only the following valid abbreviations:

["word", "1ord", "w1rd", "wo1d", "wor1", "2rd", "w2d", "wo2", "1o1d", "1or1", "w1r1", "1o2", "2r1", "3d", "w3", "4"]

Notice that only the above abbreviations are valid abbreviations of the string "word". Any other string is not a valid abbreviation of "word".

Note:

Assume s contains only lowercase letters and abbr contains only lowercase letters and digits.

Example 1:

Input

internationalization
i12iz4n

Output

true

Explanation

Given **s** = "internationalization", **abbr** = "i12iz4n":

Return true.

Example 2:

Input

apple
a2e

Output

false

Explanation

Given **s** = "apple", **abbr** = "a2e":

Return false.

Answer: (penalty regime: 0 %)

```python
import sys

def valid_word_abbreviation(word: str, abbr: str) -> bool:
    i, j = 0, 0
    m, n = len(word), len(abbr)

    while i < m and j < n:
        if abbr[j].isdigit():
            # Leading zeros are not allowed
            if abbr[j] == '0':
                return False

            # Parse the full multi-digit number
            num = 0
            while j < n and abbr[j].isdigit():
                num = num * 10 + int(abbr[j])
                j += 1
            # Skip characters in the word
            i += num
        else:
            # Characters must match exactly
            if word[i] != abbr[j]:
                return False
            i += 1
            j += 1

    # Both strings must be completely processed
    return i == m and j == n

def main():
    input_data = sys.stdin.read().split()
    if len(input_data) >= 2:
        word = input_data[0]
        abbr = input_data[1]
        result = valid_word_abbreviation(word, abbr)
        print("true" if result else "false")

if __name__ == '__main__':
    main()
```

	Input	Expected	Got	
✓	internationalization i12iz4n	true	true	✓
✓	apple a2e	false	false	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

Given a string S which is of the format USERNAME@DOMAIN.EXTENSION, the program must print the EXTENSION, DOMAIN, USERNAME in the reverse order.

Input Format:

The first line contains S.

Output Format:

The first line contains EXTENSION.
The second line contains DOMAIN.
The third line contains USERNAME.

Boundary Condition:

1 <= Length of S <= 100

Example Input/Output 1:

Input:

abcd@gmail.com

Output:

com
gmail
abcd

For example:

Input	Result
arvijayakumar@rajalakshmi.edu.in	edu.in rajalakshmi arvijayakumar

Answer: (penalty regime: 0 %)

```
S = input().strip()

username, rest = S.split('@')
domain, extension = rest.split('.', 1)

print(extension)
print(domain)
print(username)
```

	Input	Expected	Got	
✓	abcd@gmail.com	com gmail abcd	com gmail abcd	✓
✓	arvijayakumar@rajalakshmi.edu.in	edu.in rajalakshmi arvijayakumar	edu.in rajalakshmi arvijayakumar	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

Open brackets must be closed by the same type of brackets.

Open brackets must be closed in the correct order.

Constraints:

1 <= s.length <= 10^4

s consists of parentheses only '()[]{}'.

For example:

Test	Result
`print(ValidParenthesis("()"))`	`true`
`print(ValidParenthesis("()[]{}"))`	`true`
`print(ValidParenthesis("(]"))`	`false`

Answer: (penalty regime: 0 %)

Reset answer

```
def ValidParenthesis(s: str):
    stack = []
    bracket_map = {')': '(', '}': '{', ']': '['}

    for char in s:
        if char in bracket_map:
            if not stack or stack.pop() != bracket_map[char]:
                return "false"
        else:
            stack.append(char)

    if len(stack) == 0:
        return "true"
    return "false"
```

	Test	Expected	Got	
✓	`print(ValidParenthesis("()"))`	`true`	`true`	✓
✓	`print(ValidParenthesis("()[]{}"))`	`true`	`true`	✓
✓	`print(ValidParenthesis("(]"))`	`false`	`false`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

Sample Input 1

thistest123string

123

Sample Output 1

8

Answer: (penalty regime: 0 %)

```
string1 = input()
string2 = input()

print(string1.find(string2))
```

	Input	Expected	Got	
✓	thistest123string 123	8	8	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

A pangram is a sentence where every letter of the English alphabet appears at least once.

Given a string sentence containing only lowercase English letters, return true if sentence is a pangram, or false otherwise.

Example 1:

Input:

thequickbrownfoxjumpsoverthelazydog

Output:

true

Explanation: sentence contains at least one of every letter of the English alphabet.

Example 2:

Input:

arvijayakumar

Output: false

Constraints:

1 <= sentence.length <= 1000

sentence consists of lowercase English letters.

For example:

Test	Result
`print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))`	`true`
`print(checkPangram('arvijayakumar'))`	`false`

Answer: (penalty regime: 0 %)

Reset answer

```
def checkPangram(sentence):
    if len(set(sentence)) == 26:
        return "true"
    else:
        return "false"
```

	Test	Expected	Got	
✓	`print(checkPangram('thequickbrownfoxjumpsoverthelazydog'))`	`true`	`true`	✓
✓	`print(checkPangram('arvijayakumar'))`	`false`	`false`	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Given a string, determine if it is a palindrome, considering only alphanumeric characters and ignoring cases.

Note: For the purpose of this problem, we define empty string as valid palindrome.

Example 1:

```
Input:
A man, a plan, a canal: Panama

Output:
1
```

Example 2:

```
Input:
race a car

Output:
0
```

Constraints:

- `s` consists only of printable ASCII characters.

Answer: (penalty regime: 0 %)

```
def isPalindrome(s: str) -> int:
    filtered_chars = [char.lower() for char in s if char.isalnum()]
    return 1 if filtered_chars == filtered_chars[::-1] else 0


s = input().strip()
print(isPalindrome(s))
```

	Input	Expected	Got	
✓	A man, a plan, a canal: Panama	1	1	✓
✓	race a car	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

Consider the below words as key words and check the given input is key word or not.

keywords: {break, case, continue, default, defer, else, for, func, goto, if, map, range, return, struct, type, var}

Input format:

Take string as an input from stdin.

Output format:

Print the word is key word or not.

Example Input:

break

Output:

break is a keyword

Example Input:

IF

Output:

IF is not a keyword

For example:

Input	Result
break	break is a keyword
IF	IF is not a keyword

Answer: (penalty regime: 0 %)

```
keywords = {
    "break",
    "case",
    "continue",
    "default",
    "defer",
    "else",
    "for",
    "func",
    "goto",
    "if",
    "map",
    "range",
    "return",
    "struct",
    "type",
    "var",
}

word = input().strip()

if word in keywords:
    print(f"{word} is a keyword")
else:
    print(f"{word} is not a keyword")
```

	Input	Expected	Got	
✓	break	break is a keyword	break is a keyword	✓
✓	IF	IF is not a keyword	IF is not a keyword	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

The program must accept **N** series of keystrokes as string values as the input. The character **^** represents undo action to clear the last entered keystroke. The program must print the string typed after applying the undo operations as the output. If there are no characters in the string then print **-1** as the output.

Boundary Condition(s):

1 <= N <= 100
1 <= Length of each string <= 100

Input Format:

The first line contains the integer N.
The next N lines contain a string on each line.

Output Format:

The first N lines contain the string after applying the undo operations.

Example Input/Output 1:

Input:

3
Hey ^ goooo^^glee^
lucke^y ^charr^ms
ora^^nge^^^^

Output:

Hey google
luckycharms
-1

Answer: (penalty regime: 0 %)

```
n = int(input().strip())

for _ in range(n):
    s = input().rstrip("\r\n")
    stack = []

    for char in s:
        if char == "^":
            if stack:
                stack.pop()
        else:
            stack.append(char)

    result = "".join(stack)
    print(result if result else -1)
```

	Input	Expected	Got	
✓	3 Hey ^ goooo^^glee^ lucke^y ^charr^ms ora^^nge^^^^	Hey google luckycharms -1	Hey google luckycharms -1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.